# MATH 301: Homework 3

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# Section 2.3: q4

*Let* $S_4 := \left\{1 - \frac{(-1)^n}{n} : n \in \mathbb{N}\right\}$. *Find* $\inf S_4$ *and* $\sup S_4$.

Let $f(n) := 1 - \frac{(-1)^n}{n}$.

Consider the first few terms in the sequence:

$$\begin{aligned}
f(1) &= 1 - \frac{(-1)^1}{1} = 1 + 1 = 2 \\
f(2) &= 1 - \frac{(-1)^2}{2} = 1 - \frac{1}{2} = \frac{1}{2} \\
f(3) &= 1 - \frac{(-1)^3}{3} = 1 + \frac{1}{3} = \frac{4}{3} \\
f(4) &= 1 - \frac{(-1)^4}{4} = 1 - \frac{1}{4} = \frac{3}{4} \\
&\dots
\end{aligned}$$

We can reconstruct this sequence into two subsequences based on whether the index is odd or even. Let $K$ be the set of odd natural numbers and $J$ be the set of even natural numbers:

$$K := \{k \in \mathbb{N} : 2 \nmid k\} \quad J := \{j \in \mathbb{N} : 2 \mid j\}$$

For odd $k \in K$, $(-1)^k = -1$, giving the function $g(k)$:

$$g(k) = 1 + \frac{1}{k}$$

For even $j \in J$, $(-1)^j = 1$, giving the function $h(j)$:

$$h(j) = 1 - \frac{1}{j}$$

Consider the first terms of $g(k)$ where $k \in \{1, 3, 5, \dots\}$:

$$g(1) = 2, \quad g(3) = \frac{4}{3}, \quad g(5) = \frac{6}{5}, \quad \dots$$

Observe that $g(k)$ is strictly decreasing. For $x \in K$:

$$\begin{aligned}
x &< x + 2 \\
\implies \frac{1}{x} &> \frac{1}{x+2} \\
\implies 1 + \frac{1}{x} &> 1 + \frac{1}{x+2} \\
\implies g(x) &> g(x+2)
\end{aligned}$$

Consider the first terms of $h(j)$ where $j \in \{2, 4, 6, \dots\}$:

$$h(2) = \frac{1}{2}, \quad h(4) = \frac{3}{4}, \quad h(6) = \frac{5}{6}, \quad \dots$$

Observe that $h(j)$ is strictly increasing. For $z \in J$:

$$\begin{aligned}
& z < z+2 \\
\implies & \frac{1}{z} > \frac{1}{z+2} \\
\implies & -\frac{1}{z} < -\frac{1}{z+2} \\
\implies & 1-\frac{1}{z} < 1-\frac{1}{z+2} \\
\implies & h(z) < h(z+2)
\end{aligned}$$

Because $g(k)$ is strictly decreasing, its supremum is its first term (2). Because $h(j)$ is strictly increasing, its infimum is its first term ($\frac{1}{2}$). Therefore, for the entire set $S_4$:

$$\therefore \sup S_4 = 2, \quad \inf S_4 = \frac{1}{2}$$

## Section 2.3: q5

> *Find the infimum and supremum, if they exist, of each of the following sets.*
>
> (a) $A := \{x \in \mathbb{R} : 2x+5 > 0\}$, (b) $B := \{x \in \mathbb{R} : x+2 \geq x^2\}$,
> (c) $C := \{x \in \mathbb{R} : x < 1/x\}$, (d) $D := \{x \in \mathbb{R} : x^2 - 2x - 5 < 0\}$.

***Part (a):*** $A := \{x \in \mathbb{R} : 2x+5 > 0\} \implies \inf A = \frac{-5}{2},\ \nexists \sup A$
***Part (b):*** $B := \{x \in \mathbb{R} : x+2 \geq x^2\} \implies \inf B = -1,\ \sup B = 2$
***Part (c):*** $C := \{x \in \mathbb{R} : x < 1/x\} \implies \nexists \inf B,\ \sup C = 1$
***Part (d):*** $D := \{x \in \mathbb{R} : x^2 - 2x - 5 < 0\} \implies \inf D = 1-\sqrt{6},\ \sup D = 1+\sqrt{6}$

## Section 2.3: q6

> *Let* $S \subseteq \mathbb{R}, S \neq \emptyset$ *and bounded below. Prove that* $\inf S = -\sup\{-s : s \in S\}$

$$S \text{ is bounded below} \implies \exists\ u \text{ s.t. } u := \inf S$$

$$u := \inf S \implies \forall\ s \in S,\ (u \leq s) \wedge (u \text{ is the largest lowerbound})$$

$$u \text{ is the largest lowerbound} \implies \text{If } w \text{ is a lowerbound. Then } w \leq u$$

$$\text{Let } |S| = C, C \subseteq \mathbb{N} \text{ Consider ordering of } S := \{s_1 \leq s_2 \leq ... \leq s_n \leq ...\}$$

$$\text{Define } T := \{-s : s \in S\} \text{ Observe } u := \inf S \implies u \leq s_1$$

$$\text{Consider ordering of } T := \{-s_1 \geq -s_2 \geq ... \geq -s_n \geq ...\}$$

$$S \text{ is bounded below} \implies -S \text{ is bounded above} \implies \exists \sup -S$$

$$\text{Observe } u := \text{ a lower bound } S \implies -u \geq -s_1$$

$$\text{Thus } -u \geq -s_1 \geq -s_2 \geq ... \geq -s_n \geq ... \implies \forall\ x \in -S, -u \geq x \implies -u \text{ is an upperbound of } S$$

$$\text{Assume } \exists\ (z < u) \text{ s.t. } \forall\ x \in -S, z \geq x \qquad z := (-u + \frac{1}{n}),\ n \in \mathbb{N}$$

$$\implies -u + \frac{1}{n} > -s_1 \implies u - \frac{1}{n} < s_1$$

$$\implies \sup S = u$$

## Section 2.3: q8

*Let $S \subseteq \mathbb{R}, S \neq \emptyset$. Show $u \in \mathbb{R}$ is an upper bound of $S \iff (t \in \mathbb{R}) \wedge (t > u) \implies (t \notin S)$*

***Prove:*** *$u \in \mathbb{R}$ is an upper bound of $S \implies ((t \in \mathbb{R}) \wedge (t > u) \implies (t \notin S))$*

$$\text{Suppose for the sake of contracdiction: } (t \in S) \wedge (t \in \mathbb{R}) \wedge (t > u)$$

$$u \in \mathbb{R} \text{ is an upper bound of } S \implies \forall\ s \in S, u \geq s$$

$$(t \in S) \wedge (t > u) \implies u \text{ is not an upperbound of } S\ \lightning$$

$$\therefore\ (u \in \mathbb{R} \text{ is an upper bound of } S) \implies ((t \in \mathbb{R}) \wedge (t > u) \implies (t \notin S))$$

***Prove:*** *$((t \in \mathbb{R}) \wedge (t > u) \implies (t \notin S)) \implies (u \in \mathbb{R}$ is an upper bound of $S)$*

$$\text{Consider the contrapositive:}$$
$$(u \in \mathbb{R} \text{ is NOT an upper bound of } S) \implies \exists\ t \text{ s.t. } ((t \in \mathbb{R}) \wedge (t > u) \wedge (t \in S))$$

If $u \in \mathbb{R}$ is NOT an upper bound of $S \implies \exists\, t \in (S \cup \mathbb{R}) \ni t > u \implies ((t \in \mathbb{R}) \wedge (t > u) \wedge (t \in S))$

## Section 2.3: q10

*Show that if $A$ and $B$ are bounded subsets or $\mathbb{R}$. Then $A \cup B$ is a bounded set.*
*Show that $\sup(A \cup B) = \sup\{\sup A, \sup B\}$*

Let $\quad u := \sup S, \quad v := \sup B, \quad w := \sup\{u, v\}$

$(x \in A \implies x \leq u \leq w) \wedge (x \in B \implies x \leq v \leq w)$

$\implies w$ is an upper bound of $A \cup B$

$z :=$ any upper bound of $A \cup B \implies z$ is an upper bound of $A$ and $B$

$\implies u \leq z \wedge v \leq z \implies w \leq z$

$\therefore w = \sup(A \cup B)$

## Section 2.4: q2

*If $S := \{\frac{1}{n} - \frac{1}{m} | n, m \in \mathbb{N}\}$ find $\inf S$ and $\sup S$*

***Lemma 1:*** *Consider the subproblem* $\lim_{n\to\infty} \frac{1}{n}$ *where* $n$ *is an arbitrary value* $n \in \mathbb{N}$

*By backwards search:*

$$\begin{aligned}
&\forall\ \epsilon > 0, \exists\ N(\epsilon) \in \mathbb{N} \\
&\forall\ n \geq N(\epsilon), |\frac{1}{n} - 0| < \epsilon \\
&\equiv \frac{1}{n} < \epsilon \\
&\equiv n > \frac{1}{\epsilon}\ \textit{if}\ n \geq N(\epsilon) \\
&\leftarrow N(\epsilon) > \frac{1}{\epsilon} \\
&\leftarrow \textit{Archimedean Property}
\end{aligned}$$

*Forward Logic:*

$$\begin{aligned}
&\textit{Observe}\ \frac{1}{\epsilon} \in \mathbb{R} \\
&\implies \textit{Archimedean Property Holds} \\
&\implies \exists\ N(\epsilon) \in \mathbb{N}\ \textit{s.t.}\ N(\epsilon) > \frac{1}{\epsilon} \\
&\implies \forall\ n \in N(\epsilon), n > \frac{1}{\epsilon} \\
&\implies \frac{1}{n} < \epsilon \implies |\frac{1}{n} - 0| < \epsilon \\
&\therefore \frac{1}{n} \to 0\ \textit{as}\ n \to \infty
\end{aligned}$$

Thus by (Lemma 1) for an arbitrary natural number, we know $\lim_{n\to\infty} \frac{1}{n} = 0$

***Lemma 2:*** *Consider* $\frac{1}{n}$ *is strictly decreasing.*

$$\begin{aligned}
P(n) &:= \frac{1}{n} > \frac{1}{n+1} \\
P(1) &:= \frac{1}{1} > \frac{1}{2}\ \textit{holds} \\
\textit{Suppose}\ P(k) &:= \frac{1}{k} > \frac{1}{k+1} \\
&\implies k < k+1 \\
&\implies k+1 < k+2 \implies \frac{1}{k+1} > \frac{1}{k+2}
\end{aligned}$$

*Therefore, we can say* $\frac{1}{n}$ *is strictly decreasing.*

Thus by (Lemma 2) $\forall\ A := \{\frac{1}{a} : a \in \mathbb{N}\}, \sup(A) = 1$

Since $n, m \in \mathbb{N}$, we know $\frac{1}{n} \leq 1$ and $-\frac{1}{m} < 0$. Therefore, $\frac{1}{n} - \frac{1}{m} < 1$, making 1 an upper bound.

Apply the same logic to qualify $-1$ as a lower bound.

In summary:

$$n \in \mathbb{N} \implies 0 < \frac{1}{n} \leq 1$$
$$m \in \mathbb{N} \implies -1 \leq -\frac{1}{m} < 0$$

### Finding Supremum and Infimum

Find the maximum value of $\frac{1}{n}$ and minimum $\frac{1}{m}$.

$$\text{Considering the bounds (Lemma 1 \& Lemma 2):} \begin{cases} \implies \dfrac{1}{n} = 1 \\ \implies -\dfrac{1}{m} \approx 0 \end{cases}$$

Consider: $\lim_{m\to\infty}(1 - \frac{1}{m})$

$$|(1 - \frac{1}{m}) - 1| < \epsilon \iff |-\frac{1}{m} + (1-1)| < \epsilon \iff |-\frac{1}{m}| < \epsilon \iff |\frac{1}{m}| < \epsilon \iff \frac{|1|}{m} < \epsilon$$
$$\leftarrow m > \frac{1}{\epsilon} \leftarrow \text{Archimedian Property} \leftarrow \frac{1}{\epsilon} \in \mathbb{R}$$
$$\therefore \lim_{m\to\infty}(1 - \frac{1}{m}) = 1$$

Consider: $\lim_{n\to\infty}(\frac{1}{n} - 1)$

$$|(\frac{1}{n} - 1) - (-1)| < \epsilon \iff |\frac{1}{n} + (1-1)| < \epsilon \iff |\frac{1}{n}| < \epsilon \iff \frac{|1|}{n} < \epsilon \iff n > \frac{1}{\epsilon}$$
$$\leftarrow \text{Archimedian Property} \leftarrow \frac{1}{\epsilon} \in \mathbb{R}$$
$$\therefore \lim_{n\to\infty}(\frac{1}{n} - 1) = -1$$

Therefore we can say the $\sup S = 1$ and $\inf S = -1$

## Section 2.4: q7

*Let $A, B$ be bounded nonempty subsests of $\mathbb{R}$, and let $A + B := \{a + b | a \in A, b \in B\}$.*
*Prove $\sup(A+B) = \sup A + \sup B$ and $\inf(A+B) = \inf A + \inf B$.*

Consider for the following:

$$\sup A = a_s \implies \forall\ a \in A, a < a_s$$
$$\sup B = b_s \implies \forall\ b \in B, b < b_s$$

Observe:

$$\begin{array}{r} (a < a_s) \\ + \ (b < b_s) \\ \hline a + b < a_s + b_s \end{array}$$

Thus we can say $a_s + b_s$ is an upperbound. To prove it is the least upper bound, let $\epsilon > 0$. By the properties of suprema, no number strictly smaller than the supremum can be an upper bound. Therefore, there must exist some $a \in A$ and some $b \in B$ such that:

$$a > a_s - \frac{\epsilon}{2}$$

$$b > b_s - \frac{\epsilon}{2}$$

Adding these two inequalities together yields:

$$a + b > a_s + b_s - \epsilon$$

Since we found an element $a + b \in A + B$ that is strictly greater than $a_s + b_s - \epsilon$ for any arbitrary $\epsilon > 0$, no number smaller than $a_s + b_s$ can be an upper bound.

Therefore, $\sup(A + B) = a_s + b_s = \sup A + \sup B$.

Observe, $a + b$ represent arbitrary elements of $A + B$
Therefore, $\sup A + B = a_s + b_s$

$$\inf A = a_i \implies \forall\ a \in A, a > a_i$$
$$\inf B = b_i \implies \forall\ b \in B, b > b_i$$

Observe:

$$\begin{array}{r} (a > a_i) \\ +\ (b > b_i) \\ \hline a + b > a_i + b_i \end{array}$$

Thus we can say $a_i + b_i$ is a lowerbound.

Observe, $a + b$ represent arbitrary elements of $A + B$
Therefore, $\inf A + B = a_i + b_i$

To prove it is the greatest lower bound, let $\epsilon > 0$. By the properties of infima, no number strictly larger than the infimum can be a lower bound. Therefore, there must exist some $a \in A$ and some $b \in B$ such that:

$$a < a_i + \frac{\epsilon}{2}$$

$$b < b_i + \frac{\epsilon}{2}$$

Adding these two inequalities together yields:

$$a + b < a_i + b_i + \epsilon$$

Since we found an element $a + b \in A + B$ that is strictly less than $a_i + b_i + \epsilon$ for any arbitrary $\epsilon > 0$, no number larger than $a_i + b_i$ can be a lower bound.

Therefore, $\inf(A + B) = a_i + b_i = \inf A + \inf B$.

# Section 2.4: q9

*Let $X = Y := \{x \in \mathbb{R} : 0 < x < 1\}$. Define $h : X \times Y \to \mathbb{R}$ by $h(x, y) := 2x + y$.*

*(a) For each $x \in X$, find $f(x) := \sup\{h(x, y) : y \in Y\}$; then find $\inf\{f(x) : x \in X\}$.*

*(b) For each $y \in Y$, find $g(y) := \inf\{h(x, y) : x \in X\}$; then find $\sup\{g(y) : y \in Y\}$. Compare with the result found in part (a).*

### Item (a)

$$\forall\ x \in X, f(x) := \sup\{h(x, y) : y \in (0, 1)\} \implies f(x) = 2x + 1$$
$$f(x) = 2x + 1 \wedge x \in (0, 1) \implies \inf\{h(x, y) : x \in X\} = 2(0) + 1 = 1$$

### Item (b)

$$\forall\ y \in Y, g(i) := \inf\{h(x, y) : x \in (0, 1)\} \implies g(y) = 2(0) + y = y$$
$$g(y) = y \wedge y \in (0, 1) \implies \sup\{g(y) : y \in Y\} = 1$$

# Section 2.4: q14

*If $y > 0$ show $\exists\ n \in \mathbb{N} \ni \frac{1}{2^n} < y$*

Recall:

$$a > b \implies \frac{1}{a} < \frac{1}{b}$$

Observe:

$$\begin{aligned}
&P(n) : n < 2^n \\
&P(1) : 1 < 2^1 \\
&P(k) : k < 2^k \\
&2k < 2^{k+1} \\
&2k + 1 < 2k < 2^{k+1} \\
&\therefore P(n) \text{ holds } \forall\ n \in \mathbb{N}
\end{aligned}$$

Thus we can say $\frac{1}{2^n} < \frac{1}{n}$. By Archimedian Property as $y > 0 \implies \frac{1}{n} \leq y$

$$\frac{1}{2^n} < \frac{1}{n} \leq y$$

# Section 2.4: q15

*Modify the argument in Theorem 2.4.7 to show that $\exists\ y > 0 \in \mathbb{R}$ s.t. $y^2 = 3$*

> **Theorem 2.4.7** $\exists\ x > 0 \in \mathbb{R} \ni x^2 = 2$

$$S = \{s \in \mathbb{R} \mid s \geq 0 \text{ and } s^2 < 3\}$$

### Existence of Supremum

$$0^2 = 0 < 3 \implies 0 \in S \implies S \neq \emptyset$$

$$\text{Let } \forall\ s \in S, s < 3 \text{ Suppose } \exists\ s \in S \ni s \geq 3 \implies s^2 \geq 9 > 3 \ \lightning \implies \text{S is bounded above.}$$

$$S \neq \emptyset \land S \text{ is bounded above } \implies \exists\ x \in \mathbb{R} \ni x = \sup S \text{ (By Completeness Property.)}$$

### Trichotomy

By Trichotomy, one and only one, of the following must be true:

$$(x^2 < 3) \qquad (x^2 > 3) \qquad (x^2 = 3)$$

**(Case:** $(x^2 > 3)$**)**

Suppose for the sake of contracdiction $x$ is the least upper bound of $S$

$$\text{Show: } \forall\ n \in \mathbb{N}, (x - \frac{1}{n})^2 > 3 \implies \forall\ s \in S, (x - \frac{1}{n})^2 > s^2 \implies (x - \frac{1}{n}) > s$$

Observe the following:

$$(x - \frac{1}{n})^2 = x^2 - \frac{2x}{n} + \frac{1}{n^2}$$

$$x^2 - \frac{2x}{n} + \frac{1}{n^2} \xrightarrow{\textit{Dropping } \frac{1}{n^2}} (x - \frac{1}{n})^2 > x^2 - \frac{2x}{n}$$

To hold supposition, the inequality must be stricly greater than three.

$$x^2 - \frac{2x}{n} > 3$$

$$x^2 - 3 > \frac{2x}{n}$$

$$\frac{1}{x^2 - 3} < \frac{n}{2x}$$

$$\frac{2x}{x^2 - 3} < n$$

Since $x^2 > 3 \implies (x^2 - 3) > 0 \implies \exists\ n \ni x > \frac{1}{n} > 0.$

$$(x - \frac{1}{2})^2 > 3 \implies \forall\ s \in S, s^2 < 3 < (x - \frac{1}{n})^2$$

$$\implies x - \frac{1}{n} \text{ is an upper bound of } S$$

$$x - \frac{1}{n} < x \implies x \text{ is not the least upper bound of } S \ \lightning$$

$$\therefore\ (x^2 \not> 3)$$

**(Case:** $(x^2 < 3)$**)**

Observe the following:

$$(x + \frac{1}{n})^2 = x^2 + \frac{2x}{n} + \frac{1}{n^2}$$

Note $\forall\ n \in \mathbb{N}, \frac{1}{n^2} \leq \frac{1}{n}$

$$x^2 + \frac{2x}{n} + \frac{1}{n^2} \leq x^2 + \frac{2x+1}{n}$$
$$\implies x^2 + \frac{2x+1}{n} < 3$$
$$\implies \frac{2x+1}{n} < 3 - x^2$$
$$\implies \frac{n}{2x+1} > \frac{1}{3-x^2}$$
$$\implies n > \frac{2x+1}{3-x^2}$$

As $x^2 < 3 \implies 3 - x^2 > 0$. Therefore by Archimedian Property $\implies (x + \frac{1}{n}) \in S$
$\implies x$ is not an upperbound $\lightning$.

$$\therefore\ (x^2 \nless 3)$$

***As*** $x^2 \nless 3$ ***and*** $x^2 \ngtr 3$***. By Trichotomy*** $x^2 = 3$